

## Session Objectives

1. Students will be able to identify the cranial nerves for all functions (aside from olfaction, vision, and eye movements) and the particular information each nerve relays.
2. Students will be able to identify the relevant cranial nerve nuclei and their relation to the particular nerves.
3. Students will be able to identify the overall organization of the cranial nerve nuclei and each nuclei's general position.
4. Students will be able to locate injuries to these pathways given symptoms affecting specific capabilities.

# Cranial Nerve Nuclei (Motor)

| Nucleus            | nerve  | function          | effectors               |
|--------------------|--------|-------------------|-------------------------|
| abducens           | VI     | somatic motor     | eyes                    |
| oculomotor         | III    | somatic motor     | eyes                    |
| trochlear          | IV     | somatic motor     | eyes                    |
| hypoglossal        | XII    | somatic motor     | tongue                  |
| ambiguus*          | IX, XI | branchial motor   | swallowing              |
|                    | X      | branchial motor   | throat, vocals          |
| facial motor       | VII    | branchial motor   | face, upper neck        |
| spinal accessory** | XI     | branchial motor   | neck, shoulders         |
| trigeminal motor   | V      | branchial motor   | jaw, upper neck         |
| dorsal vagus       | X      | visceral efferent | gastrointestinal, heart |
| Edinger Westphal   | III    | visceral efferent | pupil constriction      |
| inferior salivary  | X      | visceral efferent | glands (mouth)          |
| superior salivary  | VII    | visceral efferent | glands (mouth, tears)   |

\* ambiguus has a brachial component and a visceral efferent component to inhibit heart rate

\*\*spinal accessory is sometimes classified as brachial and sometimes as somatic motor.

# Cranial Nerve Nuclei (Sensory)

| Nucleus             | nerve | function         | sensors                                                                   |
|---------------------|-------|------------------|---------------------------------------------------------------------------|
| gustatory           | VII   | visceral sensory | taste                                                                     |
|                     | IX, X | visceral sensory | taste                                                                     |
| solitary            | IX    | visceral sensory | blood pressure, [O2]                                                      |
|                     | X     | visceral sensory | respiratory, cardio, digestive                                            |
| trig. mesencephalic | V     | somatic sensory  | proprioception (jaw) ** <i>exception – 1<sup>st</sup> order afferents</i> |
| trig. pontine       | V     | somatic sensory  | touch/proprioception (face)                                               |
| trig. spinal        | V     | somatic sensory  | pain/temperature (face, oral cavity)                                      |
|                     | VII   | somatic sensory  | pain/temperature (ear)                                                    |
|                     | IX    | somatic sensory  | pain/temperature (throat, vocals)                                         |
|                     | X     | somatic sensory  | pain/temperature (ear, pharynx)                                           |
| vestibular          | VIII  | special sensory  | head motion / orientation                                                 |
| cochlear            | VIII  | special sensory  | hearing                                                                   |

|                           |          |                     |          |                                   |                   |
|---------------------------|----------|---------------------|----------|-----------------------------------|-------------------|
| Brainstem nuclei position | midbrain | Edinger Westphal    | III      | pupil constriction                | visceral efferent |
|                           |          | oculomotor          | III      | eyes                              | somatic motor     |
|                           |          | trig. mesencephalic | V        | jaw                               | somatic sensory   |
|                           |          | trochlear           | IV       | eyes                              | somatic motor     |
|                           | pons     | trigeminal motor    | V        | jaw, upper neck                   | branchial motor   |
|                           |          | trigeminal pontine  | V        | face                              | somatic sensory   |
|                           |          | abducens            | VI       | eyes                              | somatic motor     |
|                           |          | facial motor        | VII      | face, upper neck                  | branchial motor   |
|                           |          | superior salivary   | VII      | mouth                             | visceral efferent |
|                           |          | vestibular          | VIII     | head motion / orientation         | special sensory   |
|                           |          | cochlear            | VIII     | hearing                           | special sensory   |
|                           |          | inferior salivary   | IX       | parotid gland                     | visceral efferent |
|                           |          | solitary gustatory  | { VII    | taste                             | visceral sensory  |
|                           |          |                     | { IX, X  | taste                             | visceral sensory  |
|                           | medulla  | solitary            | { IX     | blood pressure, [O <sub>2</sub> ] | sensory           |
|                           |          |                     | { X      | respiratory, cardio, digestive    | visceral sensory  |
|                           |          | hypoglossal         | XII      | tongue                            | somatic motor     |
|                           |          | ambiguus*           | { IX, XI | swallowing                        | branchial motor   |
|                           |          |                     | { X      | throat, vocals                    | branchial motor   |
|                           |          | spinal accessory**  | XI       | neck, shoulders                   | branchial motor   |
|                           |          | dorsal vagus        | X        | gastrointestinal, heart           | visceral efferent |
|                           |          | trigeminal spinal   | { V      | face, oral cavity                 | somatic sensory   |
|                           |          |                     | { VII    | ear                               | somatic sensory   |
|                           |          |                     | { IX     | throat, vocals                    | somatic sensory   |
|                           |          |                     | { X      | ear, pharynx                      | somatic sensory   |

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|                                    |                                                                               |                                                                    |                                                         |                                                                                            |                                                                                                                                    |                                                                    |
|------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| cranial<br>nerve nuclei<br>summary | <u>SOMATIC</u><br><u>MOTOR</u>                                                | <u>BRANCHIAL</u><br><u>MOTOR</u>                                   | <u>VISCERAL</u><br><u>MOTOR</u>                         | <u>VISCERAL</u><br><u>SENSORY</u>                                                          | <u>SOMATIC</u><br><u>SENSORY</u>                                                                                                   | <u>SPECIAL</u><br><u>SENSORY</u>                                   |
| V                                  |                                                                               | trigeminal motor<br>→ jaw                                          | <div>Easy enough</div>                                  |                                                                                            | jaw proprio →<br><i>trig. mesen.</i><br>face touch & proprio →<br><i>trig. pontine</i><br>face pain & temp →<br><i>trig spinal</i> |                                                                    |
| VII                                |                                                                               | facial motor<br>→ face & stapedius                                 | sup. salivatory<br>→ tears & saliva                     | ant. tongue taste →<br><i>gustatory</i>                                                    | ear →<br><i>trig. pontine</i><br><i>trig. spinal</i>                                                                               |                                                                    |
| VIII                               | 1. The CNs are always sensory OR motor<br>2. The CNs can be sensory AND motor |                                                                    |                                                         |                                                                                            |                                                                                                                                    | orientation →<br><i>vestibular</i><br>hearing →<br><i>cochlear</i> |
| IX                                 |                                                                               | ambiguus<br>→ stylopharyngeus<br>(swallow)                         | inf. salivatory<br>→ parotid gland<br>(saliva)          | posterior tongue taste →<br><i>gustatory</i><br>carotid baroreceptors →<br><i>solitary</i> | posterior tongue,<br>pharynx and ear →<br><i>trig. pontine</i><br><i>trig. spinal</i>                                              |                                                                    |
| X                                  |                                                                               | ambiguus<br>→ pharynx&larynx<br>(vocalize, swallow, gag<br>reflex) | dorsal motor vagus →<br>“rest and digest”<br>autonomics | viscerosensory & aortic<br>baroreceptors → <i>solitary</i>                                 | ear →<br><i>trig. pontine</i><br><i>trig. spinal</i>                                                                               |                                                                    |
| XI                                 |                                                                               | spinal accessory<br>→ scm and upper<br>trapezius                   |                                                         |                                                                                            |                                                                                                                                    |                                                                    |
| XII                                | hypoglossal<br>→ tongue                                                       |                                                                    |                                                         |                                                                                            |                                                                                                                                    |                                                                    |

| <i>cranial<br/>nerve nuclei<br/>summary</i> | <u>SOMATIC<br/>MOTOR</u> | <u>BRANCHIAL<br/>MOTOR</u>                                         | <u>VISCERAL<br/>MOTOR</u>                               | <u>VISCERAL<br/>SENSORY</u>                                                                | <u>SOMATIC<br/>SENSORY</u>                                                                                                         | <u>SPECIAL<br/>SENSORY</u>                                       |
|---------------------------------------------|--------------------------|--------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>V</b>                                    |                          | trigeminal motor<br>→ jaw                                          |                                                         |                                                                                            | jaw proprio →<br><b>trig. mesen.</b><br>face touch & proprio →<br><b>trig. pontine</b><br>face pain & temp →<br><b>trig spinal</b> |                                                                  |
| <b>VII</b>                                  |                          | facial motor<br>→ face & stapedius                                 | sup. salivatory<br>→ tears & saliva                     | ant. tongue taste →<br><b>gustatory</b>                                                    | ear →<br><b>trig. pontine</b><br><b>trig. spinal</b>                                                                               |                                                                  |
| <b>VIII</b>                                 |                          |                                                                    |                                                         |                                                                                            |                                                                                                                                    | orientation→<br><b>vestibular</b><br>hearing→<br><b>cochlear</b> |
| <b>IX</b>                                   |                          | ambiguus<br>→ stylopharyngeus<br>(swallow)                         | inf. salivatory<br>→ parotid gland<br>(saliva)          | posterior tongue taste →<br><b>gustatory</b><br>carotid baroreceptors →<br><b>solitary</b> | posterior tongue,<br>pharynx and ear →<br><b>trig. pontine</b><br><b>trig spinal</b>                                               |                                                                  |
| <b>X</b>                                    |                          | ambiguus<br>→ pharynx&larynx<br>(vocalize, swallow, gag<br>reflex) | dorsal motor vagus →<br>“rest and digest”<br>autonomics | viscerosensory & aortic<br>baroreceptors → <b>solitary</b>                                 | ear →<br><b>trig. pontine</b><br><b>trig. spinal</b>                                                                               |                                                                  |
| <b>XI</b>                                   |                          | spinal accessory<br>→ scm and upper<br>trapezius                   |                                                         |                                                                                            |                                                                                                                                    |                                                                  |
| <b>XII</b>                                  | hypoglossal<br>→ tongue  |                                                                    |                                                         |                                                                                            |                                                                                                                                    |                                                                  |

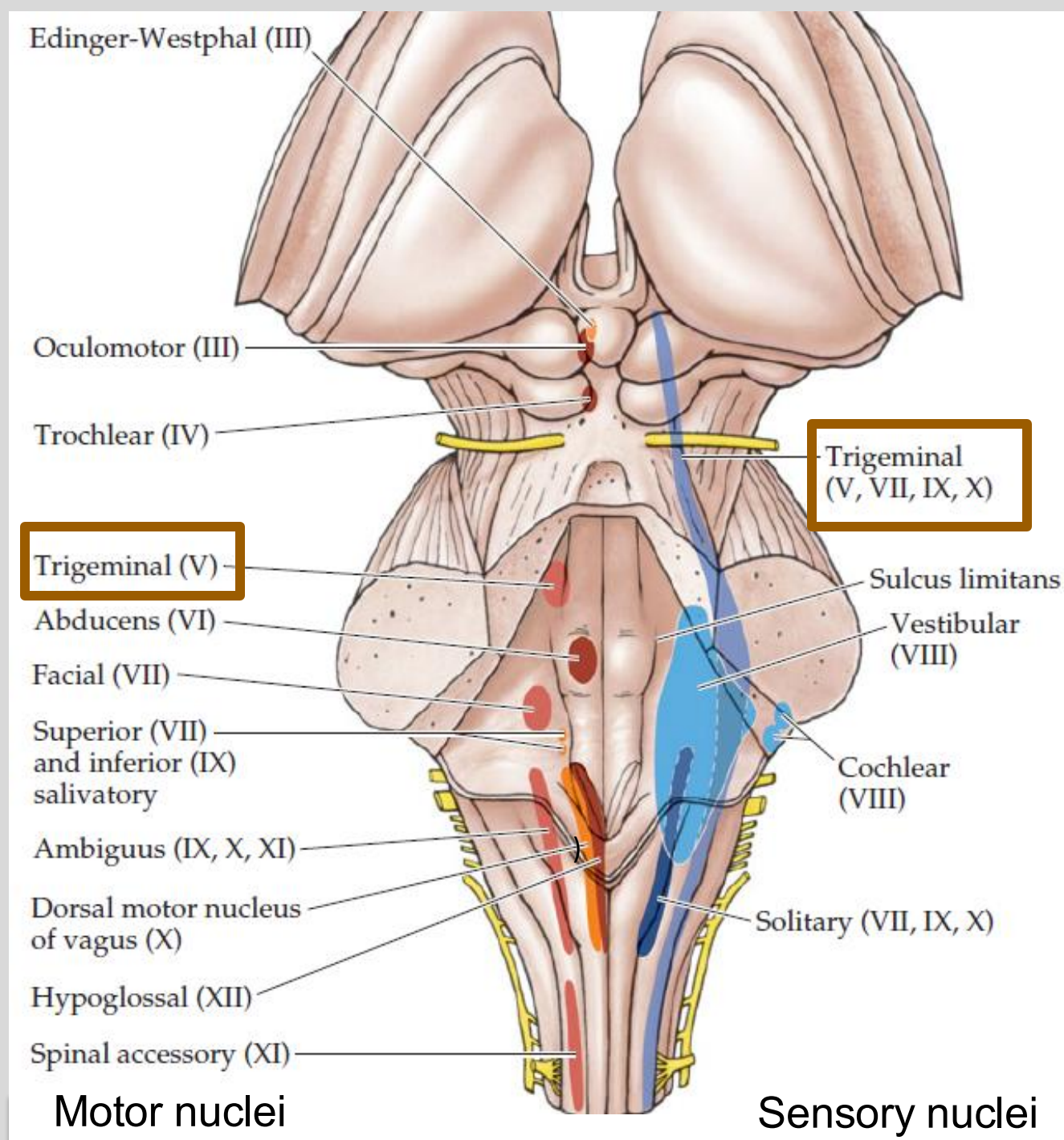
A helpful vignette

A helpful name

Tough, but one pattern



## CN V (trigeminal)



# CN V (trigeminal)

Three major divisions: V1 **ophthalmic** – somatosensation  
V2 **maxillary** – somatosensation  
V3 **mandibular** – somatosensation AND motor

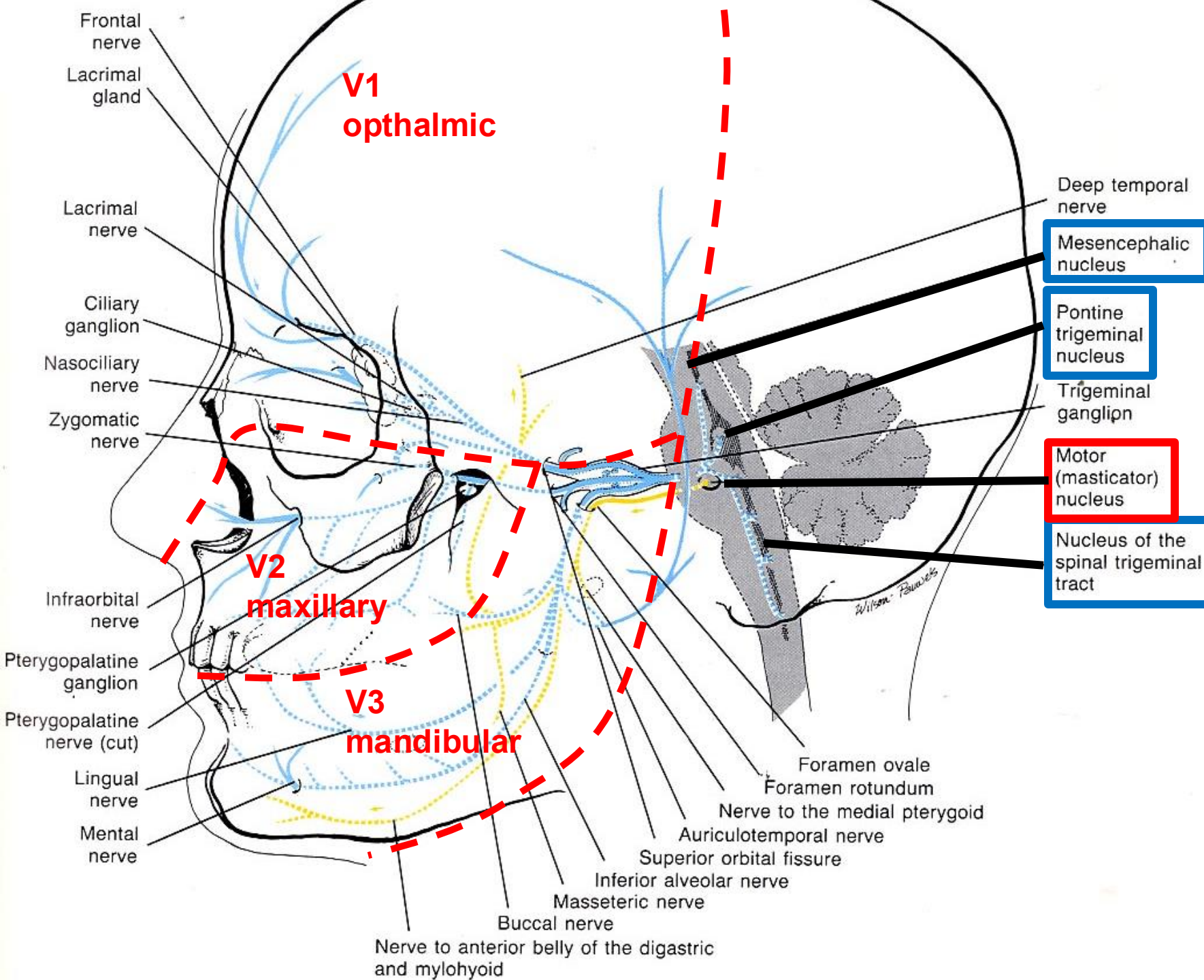
**trigeminal pontine nuclei** ← V1-3 touch & proprioception  
**trigeminal spinal nuclei** ← V1-3 pain & temperature } ipsilateral face incl. eye  
& anterior 2/3 tongue

trigeminal pontine and spinal nuclei also receive skin feedback  
from CN VII (ear), CN IX (posterior 1/3 tongue, ear, throat), and CN X (ear)

**trigeminal mesencephalic nuclei** ← V3 jaw proprioception (the nuclei is 1<sup>st</sup> order neuron)  
**trigeminal motor nuclei** → V3 muscles of mastication

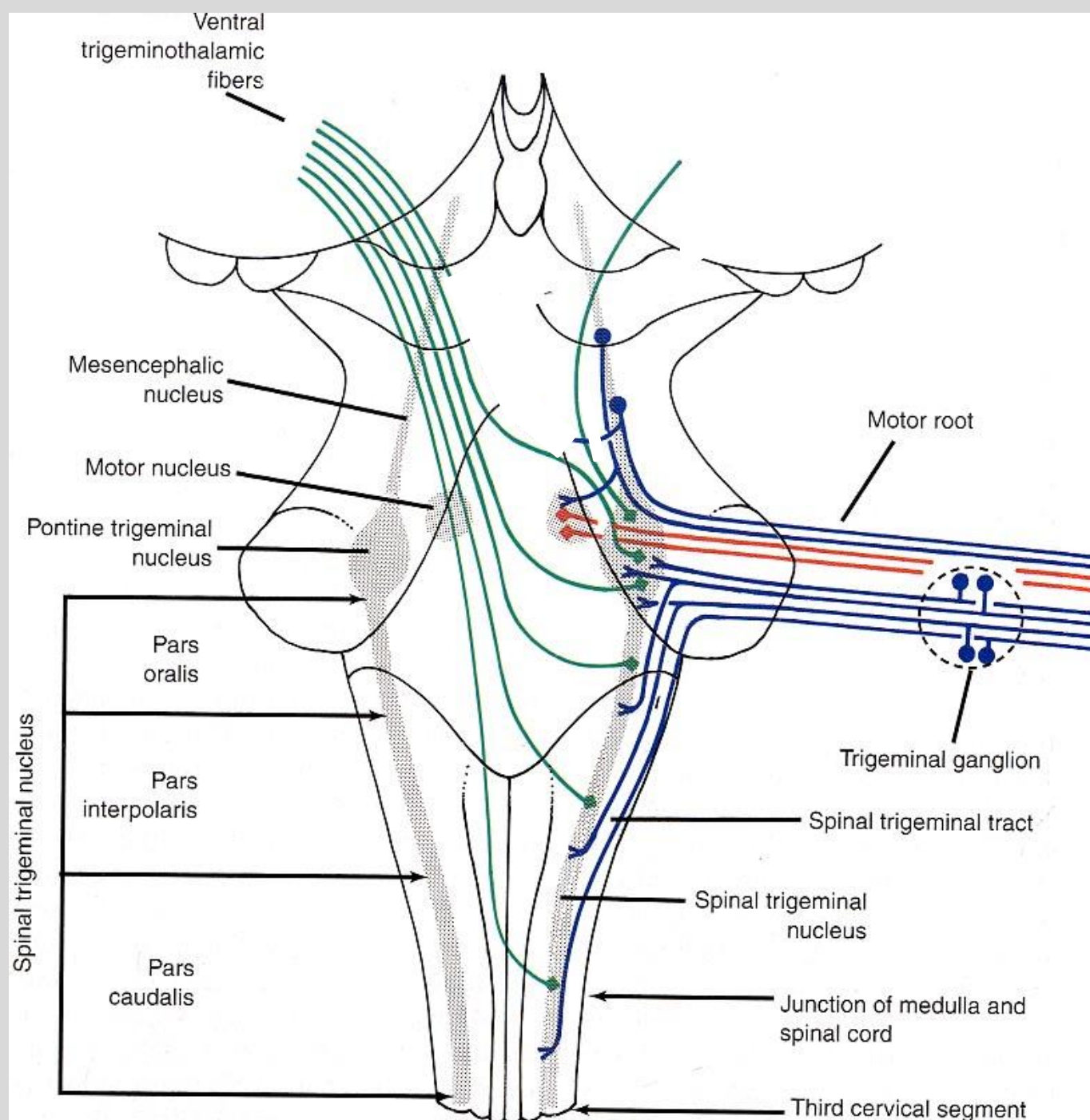


## CN V (trigeminal)



*Note: skin sense of the back of head is handled by spinal nerves*

# CN V (trigeminal)



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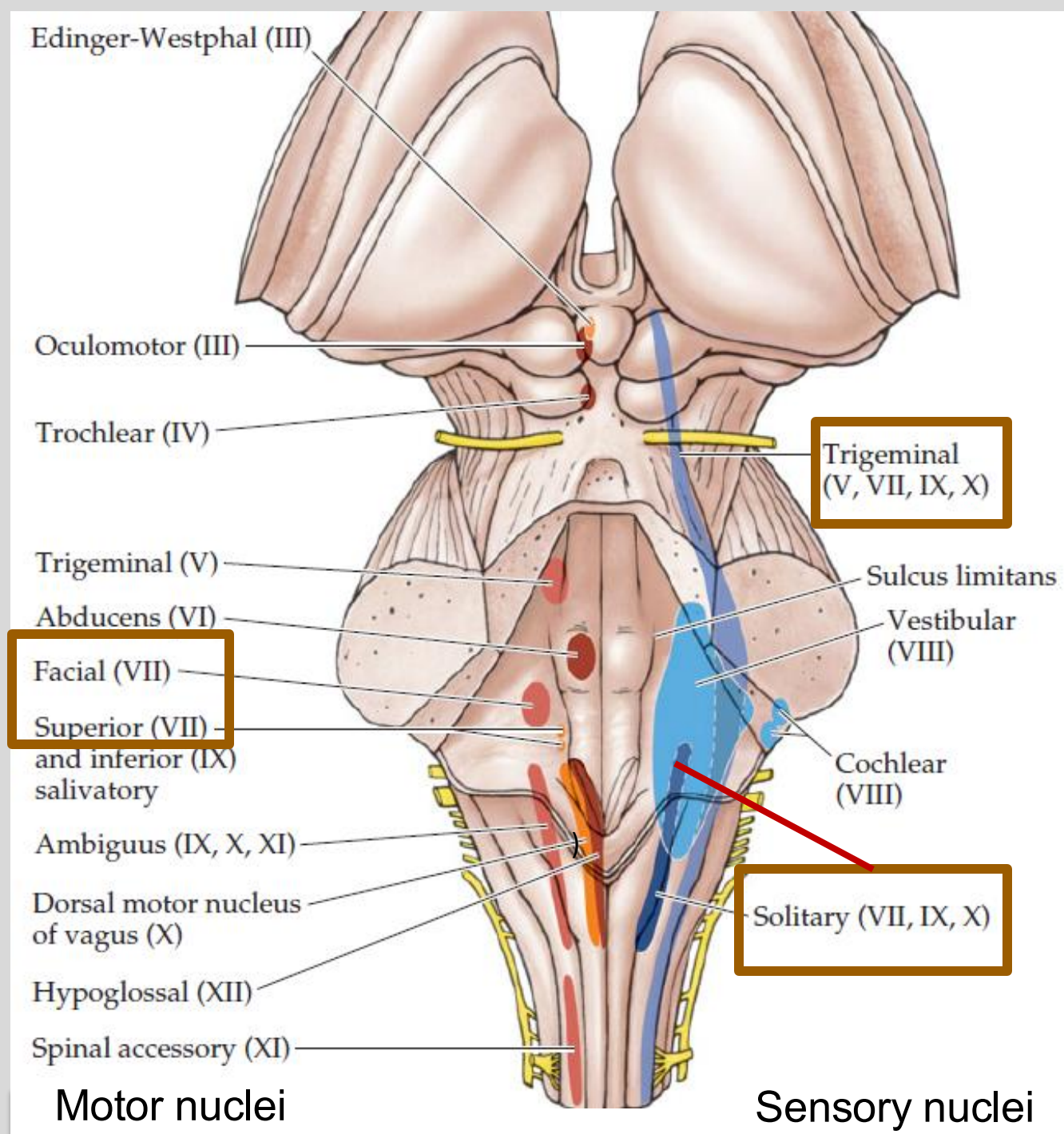
Principle and spinal trigeminal nuclei project to the **contralateral** thalamus

Pain/temp fibers descend to spinal trigeminal nucleus

Lesion at upper spinal level would blunt pain/temp for ipsilateral face but not touch or proprioception!



## CN VII (facial)



## CN VII (facial)

**facial motor nucleus** → facial expression muscles, diaphragm, stapedius (middle ear)

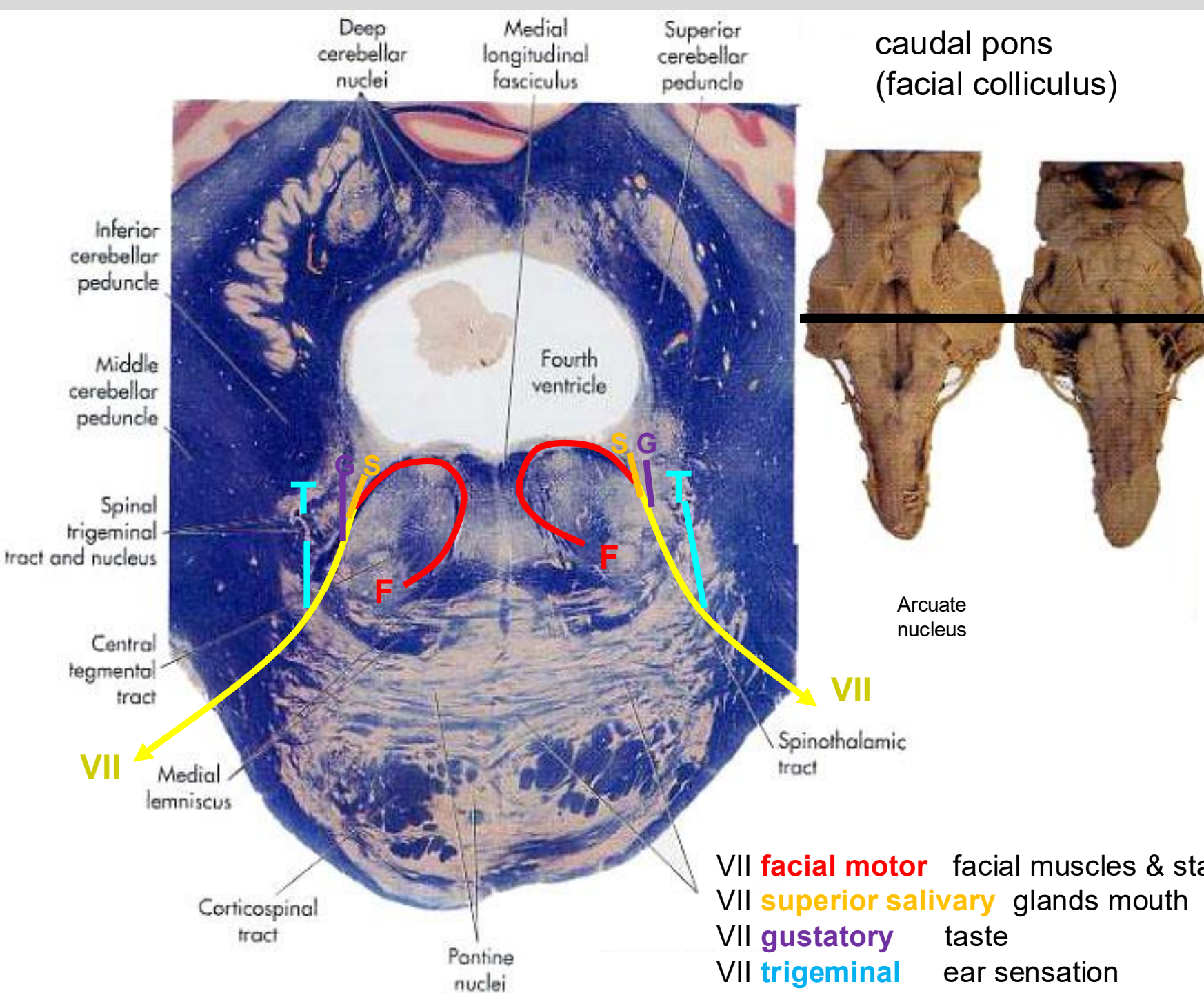
**sup. salivatory nucleus** → lacrimal (tears), submandibular and sublingual glands (saliva)

**gustatory nucleus** ← taste (anterior 2/3 of tongue)

**trigeminal nucleus** ← posterior surface of external ear skin senses

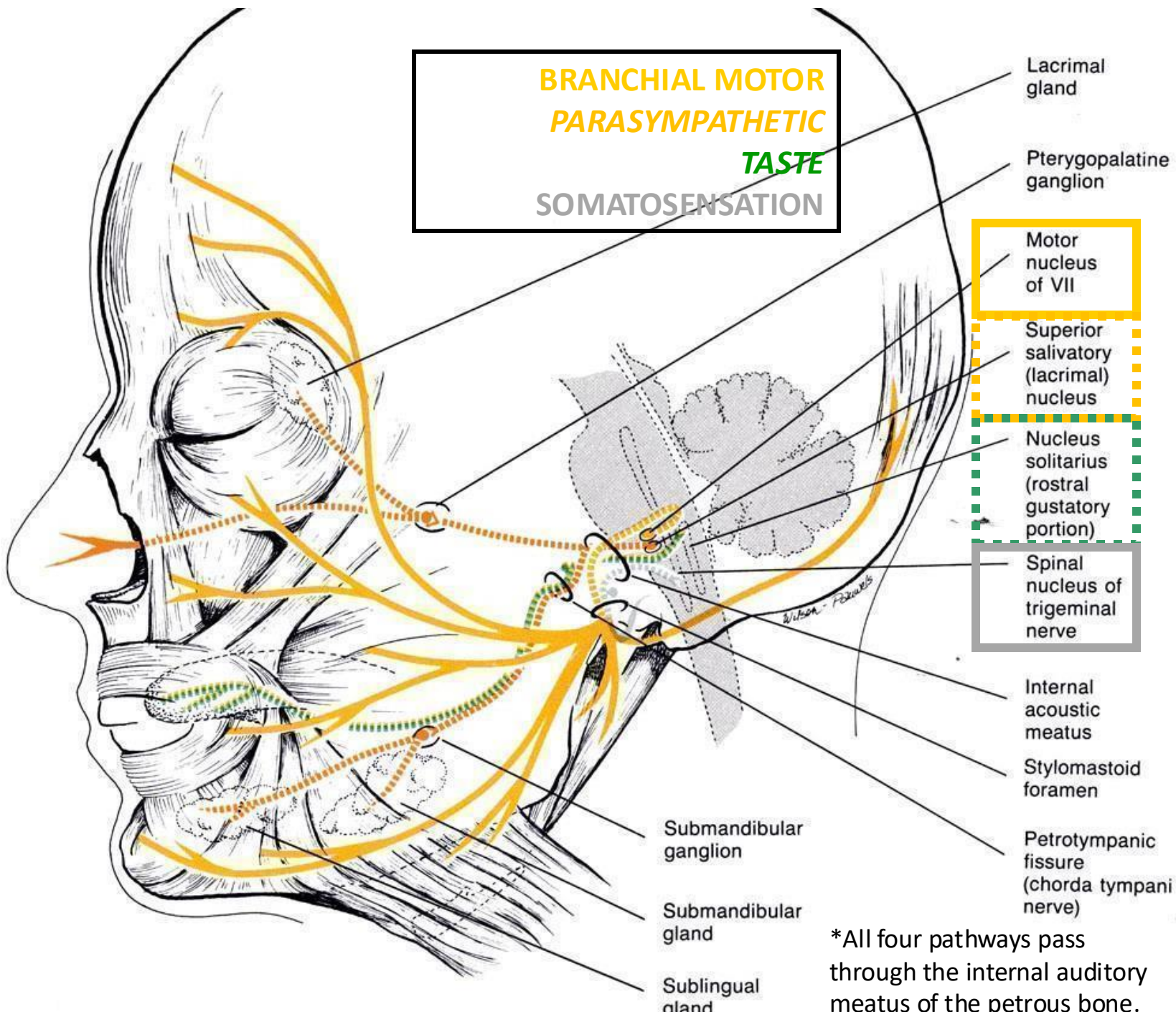
*“Enjoying a Greek olive, not chewing, just tasting and smiling and drooling a bit, and the tears well up its just so tasty!! and then my wife drags me out of the store by the ear.”*

## CN VII (facial)





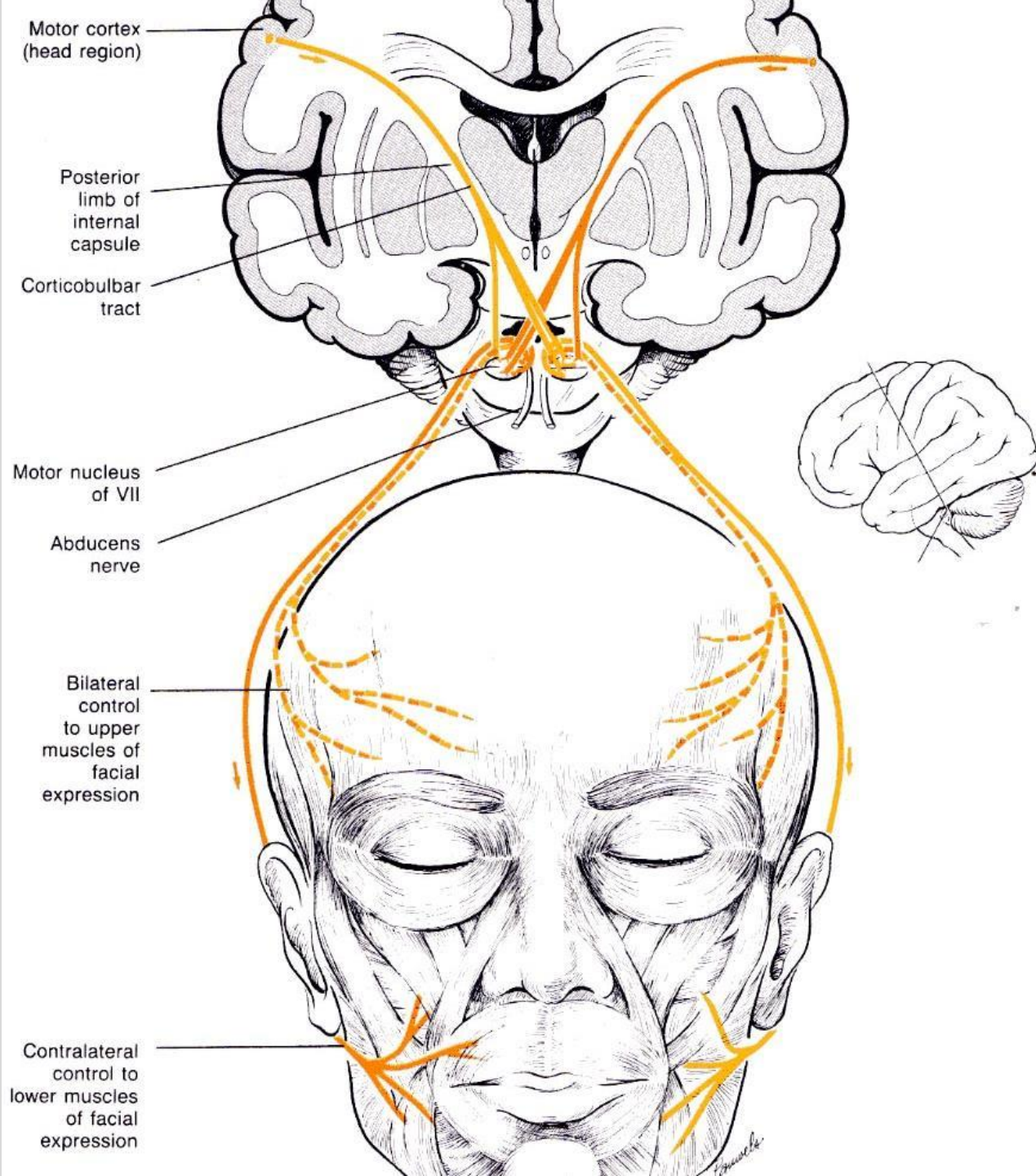
CN VII (facial)



\*All four pathways pass through the internal auditory meatus of the petrous bone.



## CN VII (facial)



*forehead / eyebrow /*

*cortical motor neurons project **BILATERALLY** to facial motor nuclei*

*Cheek cortical motor neurons project **contralaterally** to facial motor nuclei*

Edinger-Westphal (III)

Oculomotor (III)

Trochlear (IV)

Trigeminal (V)

Abducens (VI)

Facial (VII)

Superior (VII)

and inferior (IX)  
salivatory

Ambiguus (IX, X, XI)

Dorsal motor nucleus  
of vagus (X)

Hypoglossal (XII)

Spinal accessory (XI)

Motor nuclei

Trigeminal  
(V, VII, IX, X)

Sulcus limitans

Vestibular  
(VIII)

Cochlear  
(VIII)

Solitary (VII, IX, X)

Sensory nuclei

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## CN IX (glossopharyngeal)

# CN IX (glossopharyngeal)

★ **nucleus ambiguus** → *stylopharyngeus* muscle— swallowing (pharyngeal phase)

★ **inferior salivatory nucleus** → parotid gland secretion

★ **gustatory nucleus** ← posterior 1/3 of the tongue

**solitary nucleus** ← baroreceptor feedback from carotid sinus and body

★ **trigeminal nucleus** ← posterior 1/3 tongue, throat, and ear

Remember the name: **glosso-pharyngeal**

tongue/speech e.g. glossary,  
glossolalia (speaking in tongues)

pharynx



**nucleus ambiguus** → **stylopharyngeus muscle**

**inf salivatory** → **parotid gland**

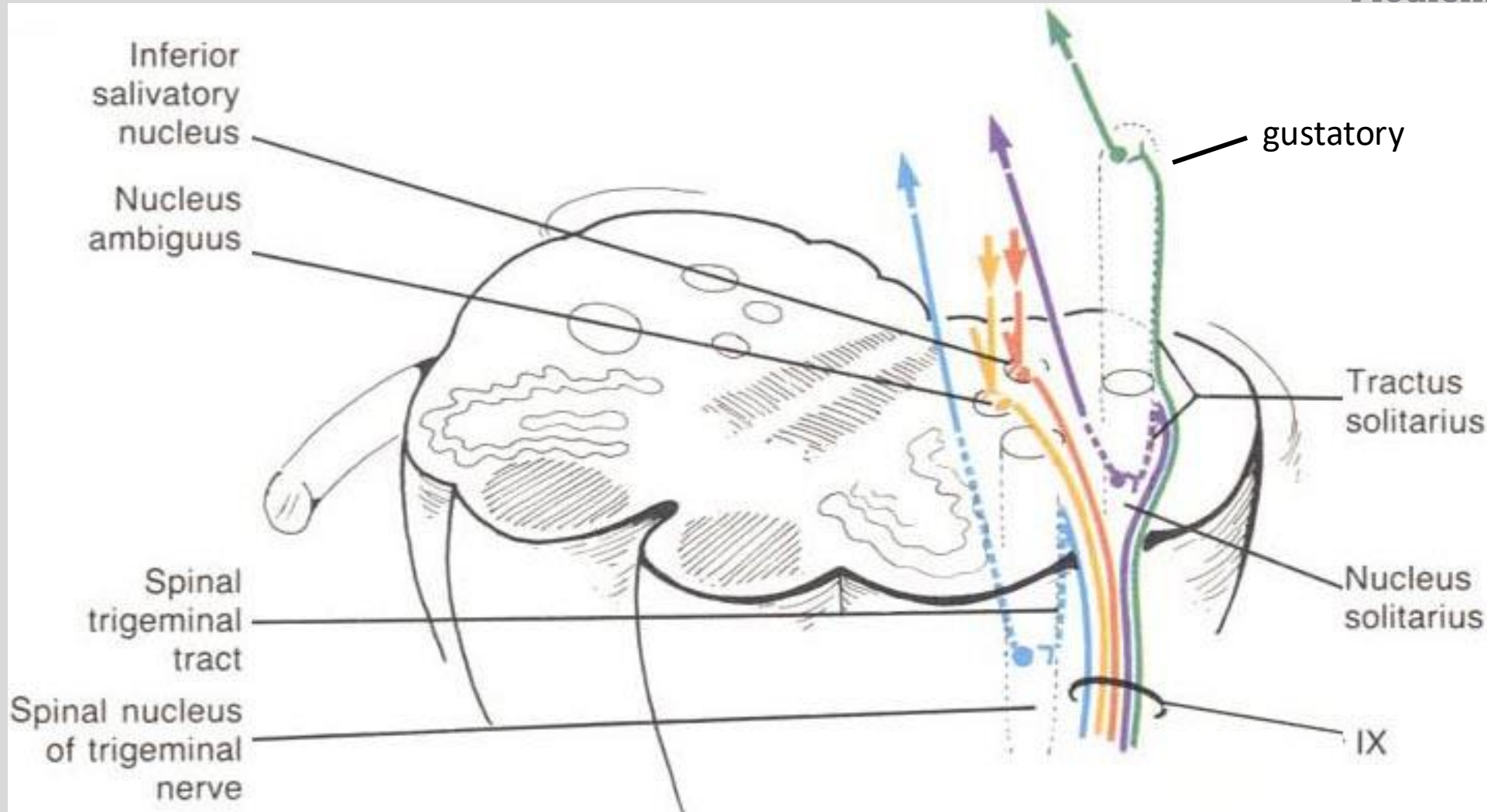
**gustatory** ← **posterior tongue** (note switch in color from other figs)

**solitary** ← **carotid sinus and body**

**trigeminal** ← **posterior tongue, throat, and ear**

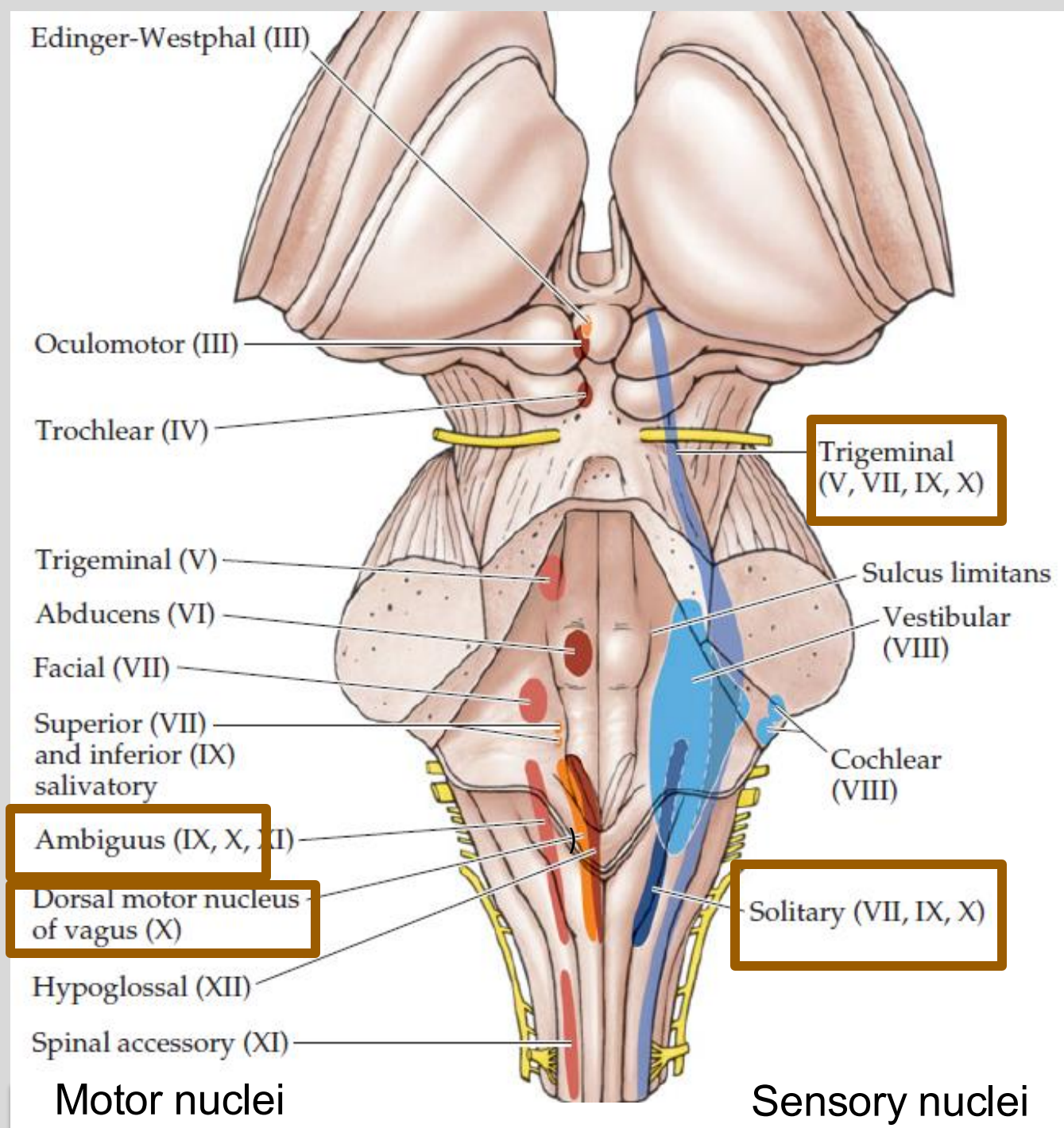
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## CN X (vagus)



# CN X (vagus)

**nucleus ambiguus** → striated muscle of soft palate, pharynx (gag reflex),  
and muscles for vocal cord in larynx

**dorsal motor nucleus** → many parasympathetic responses , “rest and digest” incl.  
slower heart rate and increased secretions and motility of the GI tract

**solitary nucleus** ← visceral sensory from larynx, esophagus, trachea, (i.e. mucous membranes),  
abdominal & thoracic viscera & aortic arch stretch receptors, aortic bodies  
(chemoreceptors) (Nucleus of Solitary Tract, NST)

**trigeminal nucleus** ← skin of the back of ear and ear canal

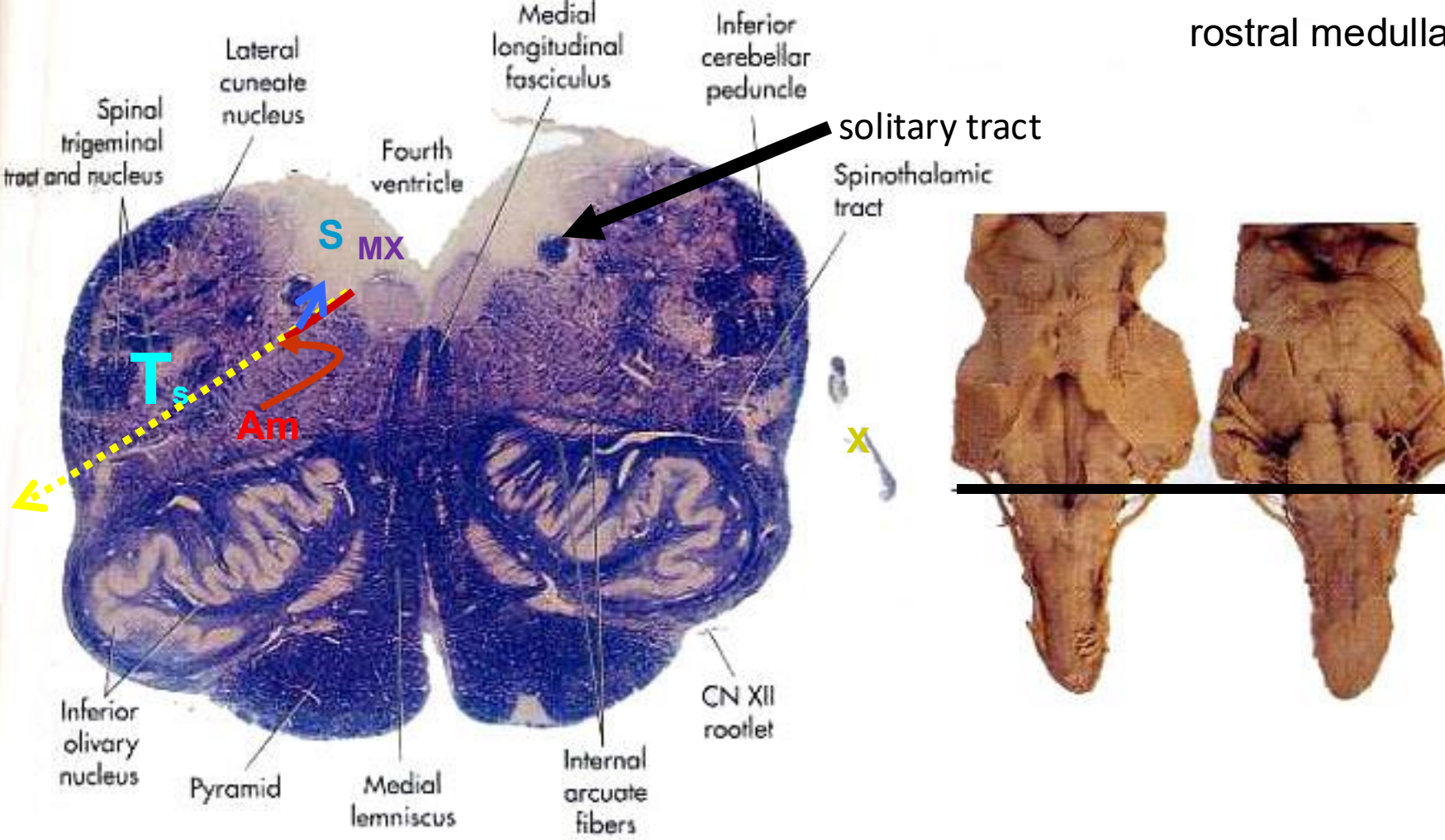
The vagus nerve is the largest conduit of autonomic output from the brainstem  
AND autonomic input to the brainstem. In fact, 65-80% are autonomic afferents!

rostral medulla

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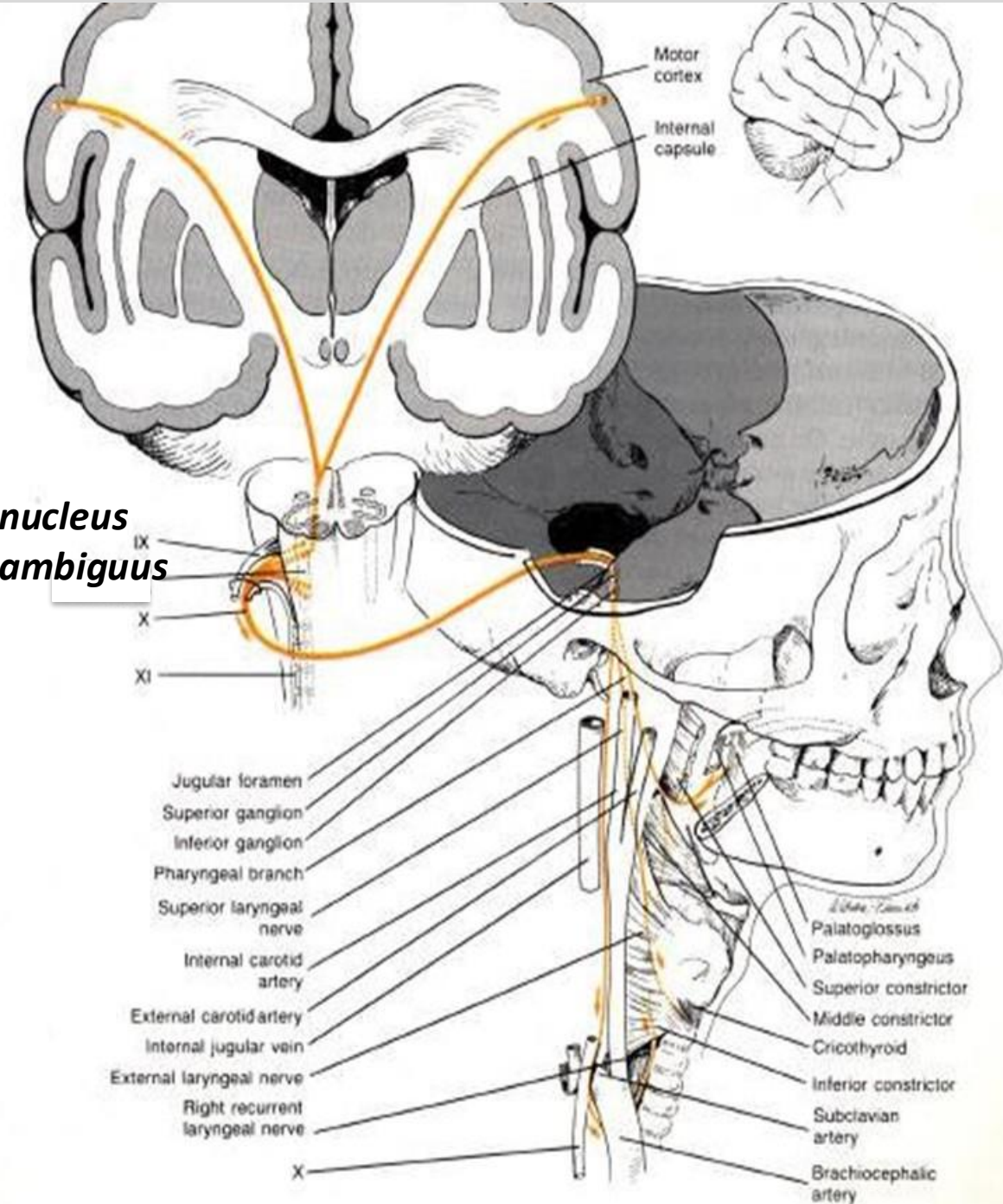
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CN X (vagus)



- |   |                           |                                                  |
|---|---------------------------|--------------------------------------------------|
| X | <b>ambiguus</b>           | throat, vocals motor, <i>and heart efferents</i> |
| X | <b>dorsal motor vagus</b> | gastrointestinal, heart efferent                 |
| X | <b>solitary</b>           | respiratory, cardio, digestive feedback          |
| X | <b>trigeminal</b>         | ear somatosensation                              |





## CN X (vagus)

nucleus ambiguus  
projects to **ipsilateral**  
throat muscle

**unilateral medulla**  
lesion will cause  
**ipsilateral** paralysis

the nucleus ambiguus  
is **bilaterally** controlled  
by upper / cortical  
motor neurons

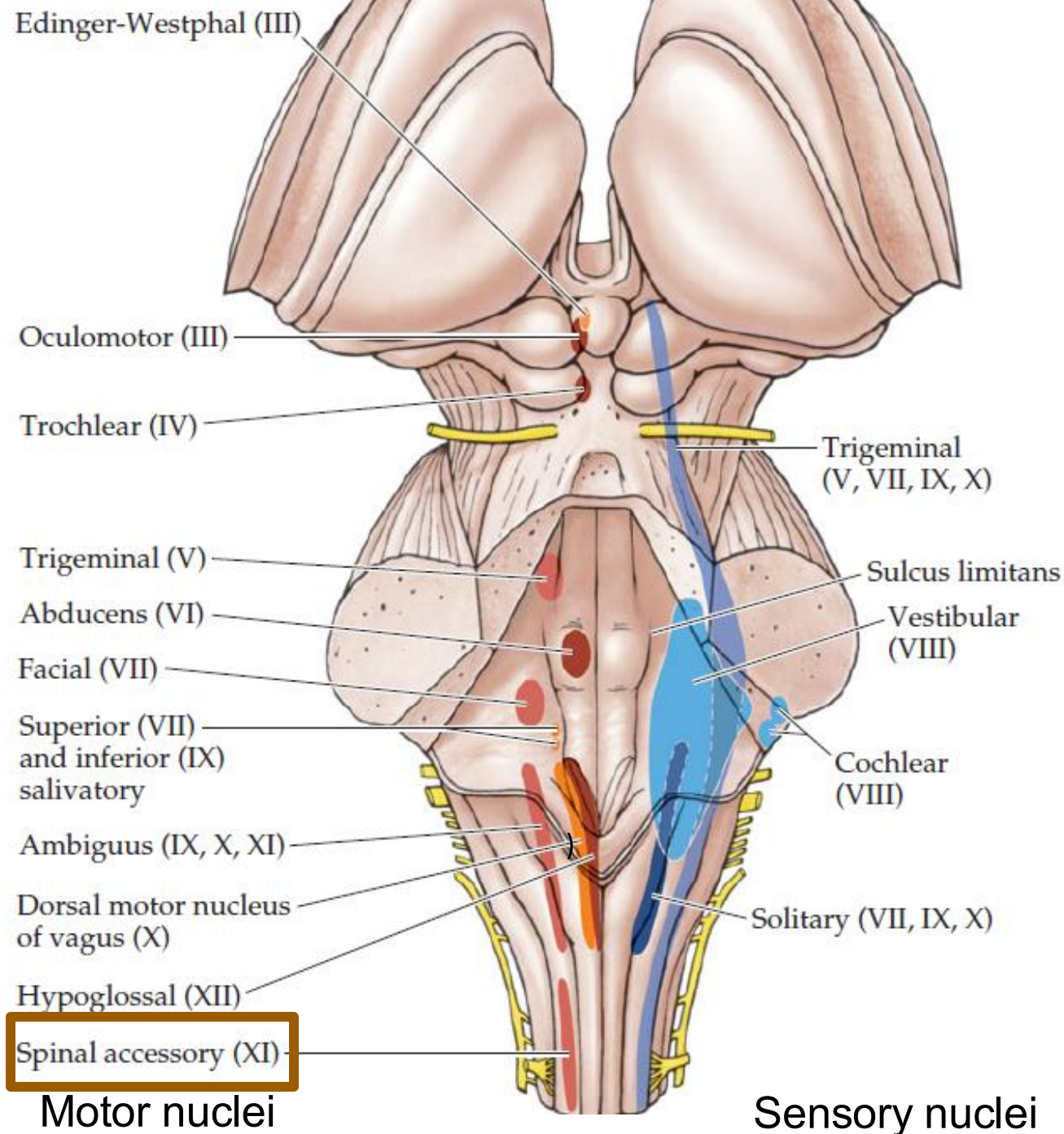
**unilateral cortical**  
lesion will be largely  
compensated

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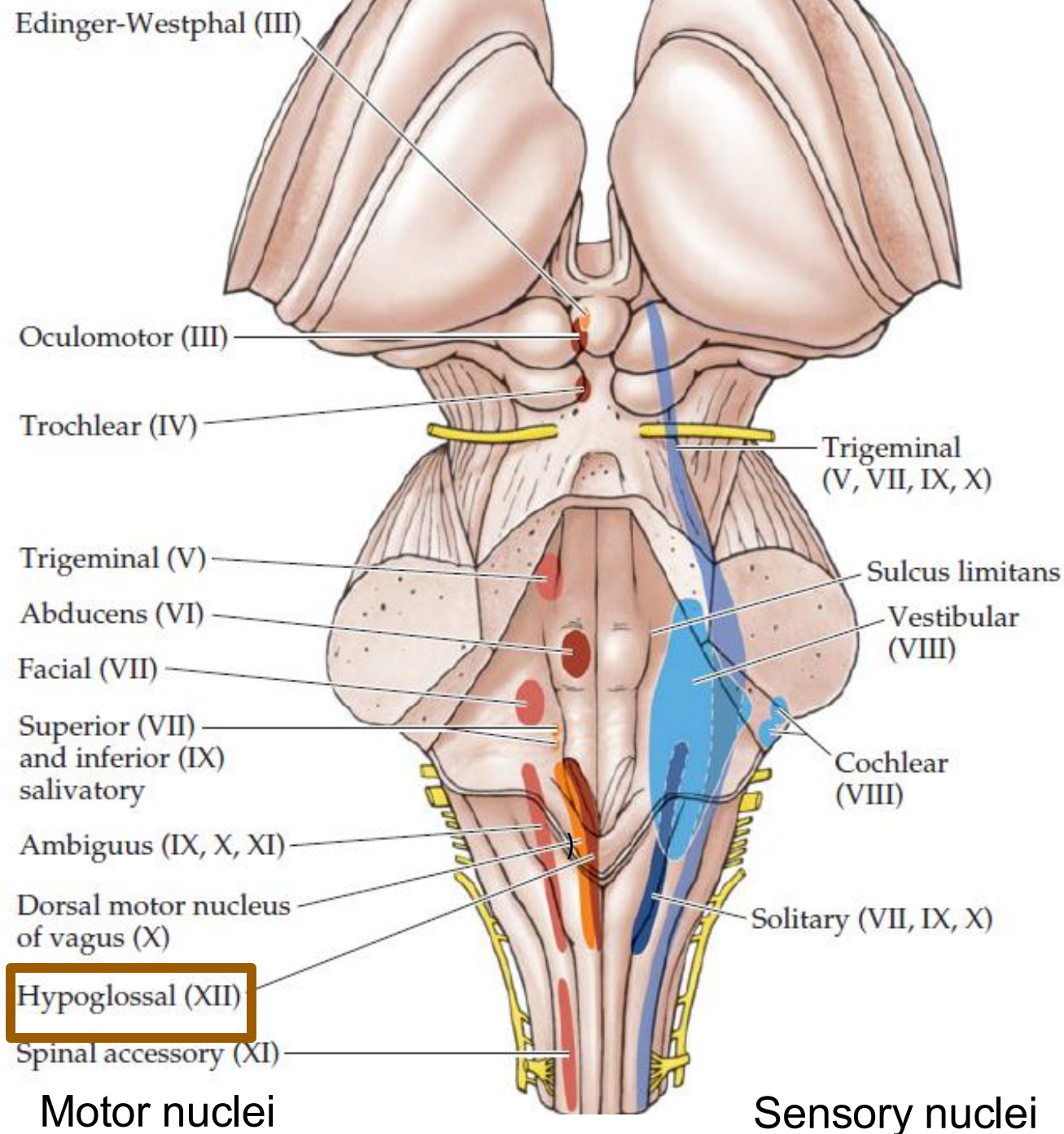
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## CN XI (spinal accessory)



## CN XII (hypoglossal)



## CN XII (hypoglossal)

Hypoglossal nerve palsy

Destruction of the hypoglossal nucleus, or interruption of the motor axons

Tongue deviates to the weak side  
Due to unopposed action of contralateral *genioglossus* muscle

Muscle atrophy

Isolated damage is relatively rare:  
Tumors, iatrogenic, penetrating trauma.

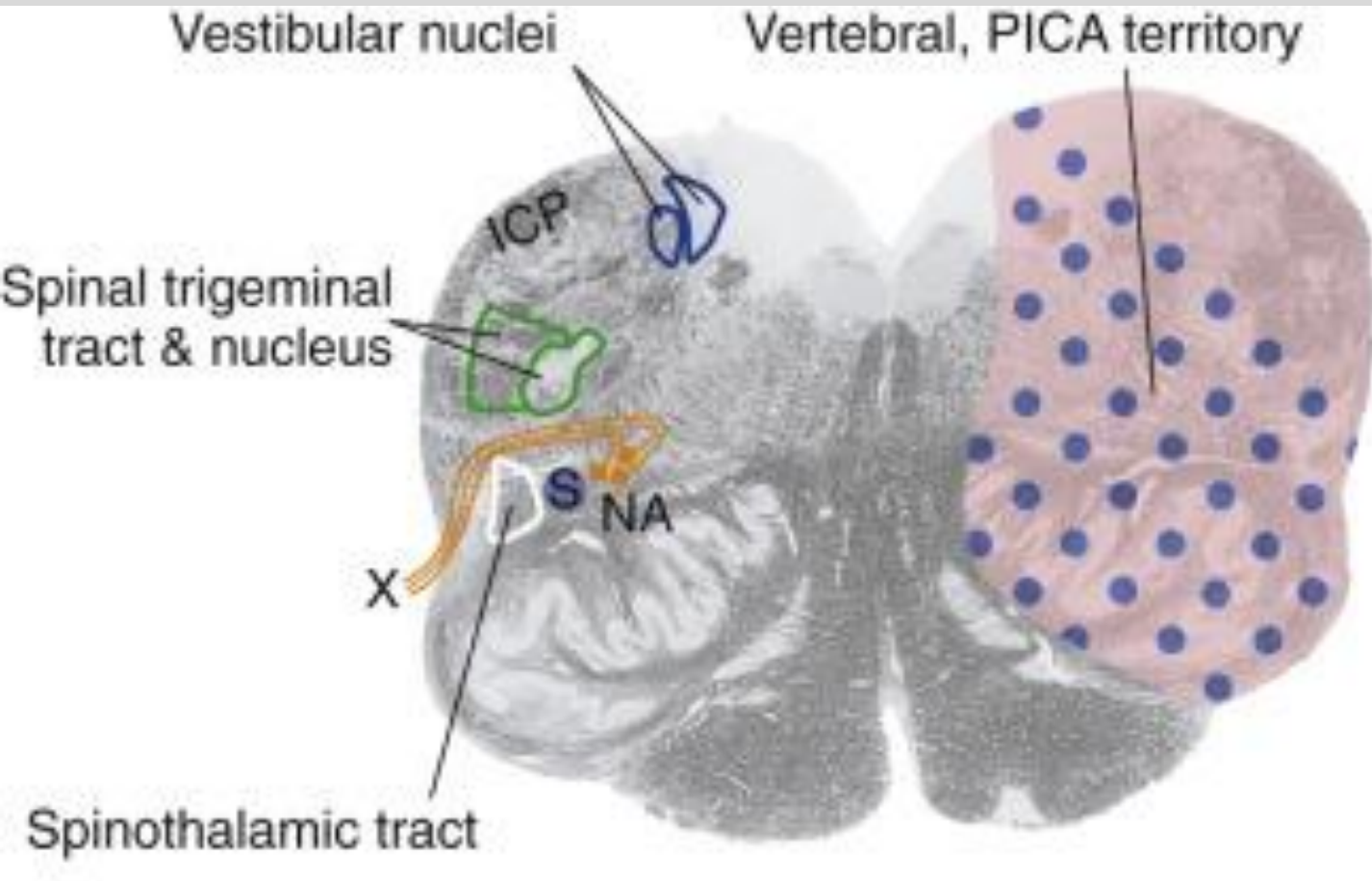




# Lateral medullary syndrome

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Loss of vascular supply  
(posterior inferior cerebellar artery)

Damage to:

Spinothalamic tract - contralateral pain and temp deficit of body

Spinothalamic nucleus (S) - ipsilateral pain and temp deficit of face

Nucleus ambiguus (NA) - hoarseness and difficulty in swallowing

Vestibular nucleus – falling to the ipsilateral side

Inferior cerebellar peduncle (ICP) –  
dyscoordination on ipsilateral leg and arm

## Summary

- There is a rational medial-lateral layout of the cranial nerve nuclei.
- The pontine trigeminal nucleus processes touch & proprioception. The spinal trigeminal nucleus processes pain & temperature. Most afferents are from the trigeminal nerve. Others from facial, glossopharyngeal, and vagus nerve.
- The facial nerve relays motor efferents from the facial motor nucleus and superior salivary nucleus. Relays afferents to gustatory nucleus and trigeminal nuclei.
- The vestibulocochlear nerves uniquely supply the afferents to the vestibular and cochlear nuclei.
- The glossopharyngeal nerve is aptly named. The ambiguous nucleus controls swallowing, the gustatory nucleus receives tongue taste, and tongue/throat somatosensation is sent to the trigeminal nucleus.
- The vagus relays autonomic input to the solitary nucleus and output from the dorsal motor vagus. It also relays pharyngeal and laryngeal control from the ambiguous nucleus.
- Hypoglossal and spinal accessory nerves are motor efferents uniquely from the hypoglossal and spinal accessory nuclei.